Umadhatri Durvasula

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Summary

Infrastructure engineer who builds backends that hold under load and cloud platforms that don't need babysitting. Cut AWS lab deployment time to under 4 minutes using Golden AMIs and eliminated race conditions in concurrent job handling. One year in — I've already shipped production systems at scale.

Experience

Infrastructure Engineer (Project Engineer) — Feb 2025 – Present
CySTAR – IIT Madras — *Chennai, India*

- Architected a production-grade, multi-tenant CyberRange platform on AWS supporting 500+ participants across fully isolated lab environments, using Headscale VPN as the sole ingress and isolation layer to eliminate cross-deployment traffic bleed
- Reduced lab provisioning time to under 4 minutes by engineering a Golden AMI pipeline with Terraform, cutting manual deployment overhead to zero across all lab lifecycle stages
- Eliminated race conditions in concurrent job handling by implementing PostgreSQL `FOR UPDATE SKIP LOCKED` worker processes, enabling safe parallel deployments across 500+ simultaneous users without contention
- Secured the full platform with JWT-based authentication backed by a Redis blocklist, ensuring stateless session invalidation across a distributed FastAPI backend
- Diagnosed and resolved a 91% failure rate in a production RAG pipeline by restructuring the knowledge base architecture and integrating LLM-based answer validation, lifting answer quality by 54% against a held-out evaluation set

Projects

CTF Challenge Design | *Binary exploitation, web security, AI difficulty scoring* — 2025

- Designed and deployed 10+ CTF challenges in web exploitation and binary analysis, reaching 1,000+ participants across two cohorts
- Built an AI-assisted difficulty calibration pipeline that validated challenge progression across beginner and advanced skill levels

Education

Sreenidhi Institute of Science and Technology — Dec 2021 – Jun 2025
Bachelor of Technology in Electronics and Communications Engineering — *GPA: 8.92/10.0*

Technical Skills

Languages & Frameworks: Python, FastAPI, Bash
Infrastructure & Cloud: AWS (EC2, VPC, IAM, S3, ECS), Terraform, Docker, Headscale VPN, PostgreSQL, Redis
Security & Binary Analysis: Ghidra, angr, CTF design, Linux, Git

Certifications

Google Professional Certificate in Cybersecurity: Security operations, incident detection & response (NIST framework), network traffic analysis, SIEM tools, vulnerability management